MARKOV ANALYTICS

Classification of States

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Understanding Markov States

Definitions and Concepts

In Markov chains, a state represents a specific condition or position within a stochastic process. States transition based on probabilities, influencing long-term behaviors and classifications within the system.

Transient vs. Recurrent States

Definitions and Distinctions

In Markov chains, **transient states** may be visited **only a few times**, while **recurrent states** are revisited infinitely often, highlighting important differences in long-term behavior and stability.

Absorbing States

Characteristics and Examples

An **absorbing state** is a unique state in a Markov chain where, once entered, it cannot be left. Such states are vital for understanding long-term behavior and system stability.

Periodic and Aperiodic States

Understanding State Characteristics

Periodic states revisit at regular intervals, while aperiodic states do not follow such patterns. Recognizing these classifications is crucial for analyzing **long-term behavior** in Markov chains.

Ergodic State

Long-Term Behavior Overview

An **ergodic state** is characterized by its ability to return to itself over time, ensuring that every state in the Markov chain can be reached from any starting point, fostering convergence.

How Are States Classified?

Explore the types of states in Markov chains